PSD study update

Yingrui Hou (LPCA)

What have been done

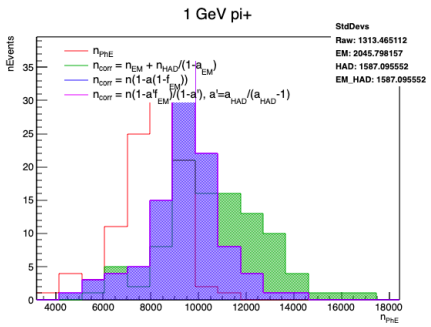
- Simulation samples: e/p/pi, into 2m x 2m x 2m ZnWO4(BGO) crystal
  - Energy range: 1 GeV - 20 GeV

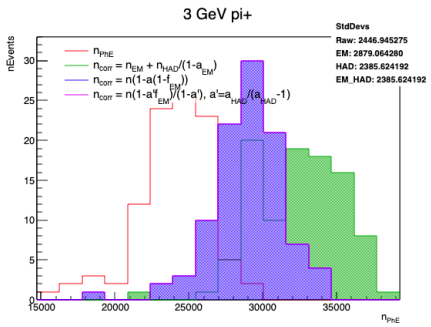
  ⇒ Correlation between deposit hadronic shower energy and total energy is validated, and independent to the input energy
- Simulation samples: pi, into 2m x 2m x 2m ZnWO4(BGO) crystal + scintillation process
  - Energy range: 1GeV – 20 GeV
  - Time constants of scintillating photons are updated dynamically based on particle type and dE/dx (pars taken from α and γ)

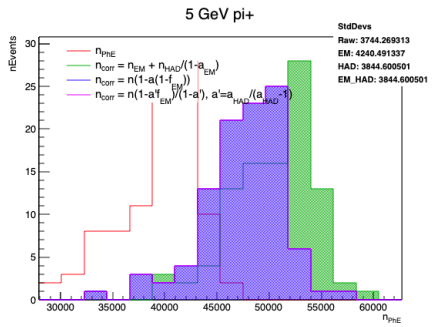
  ⇒ Correlation between fraction of deposit EM shower energy and total energy is validated

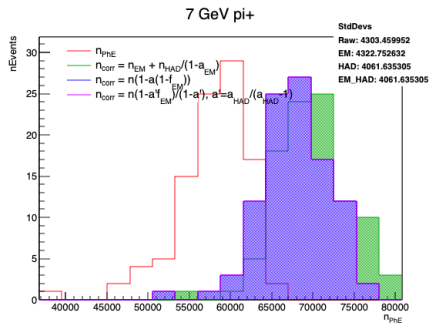
  ⇒ The energy resolutions are checked, and better resolutions are found.

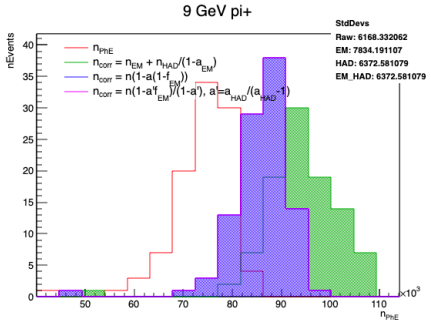
The “reconstructed” energy in samples

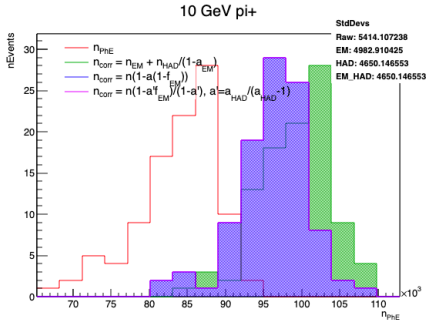


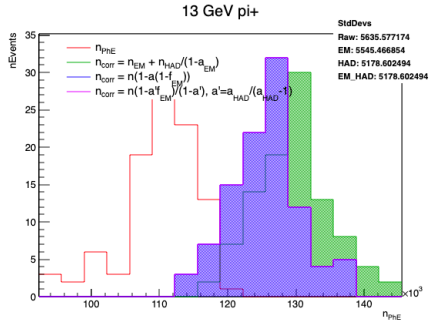


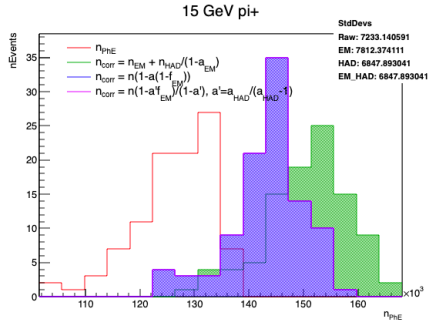


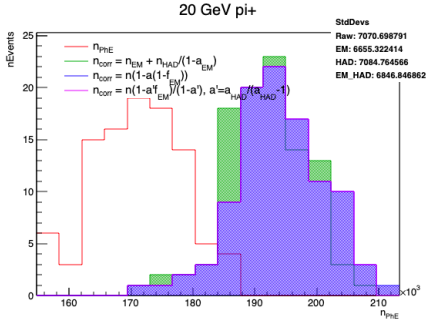




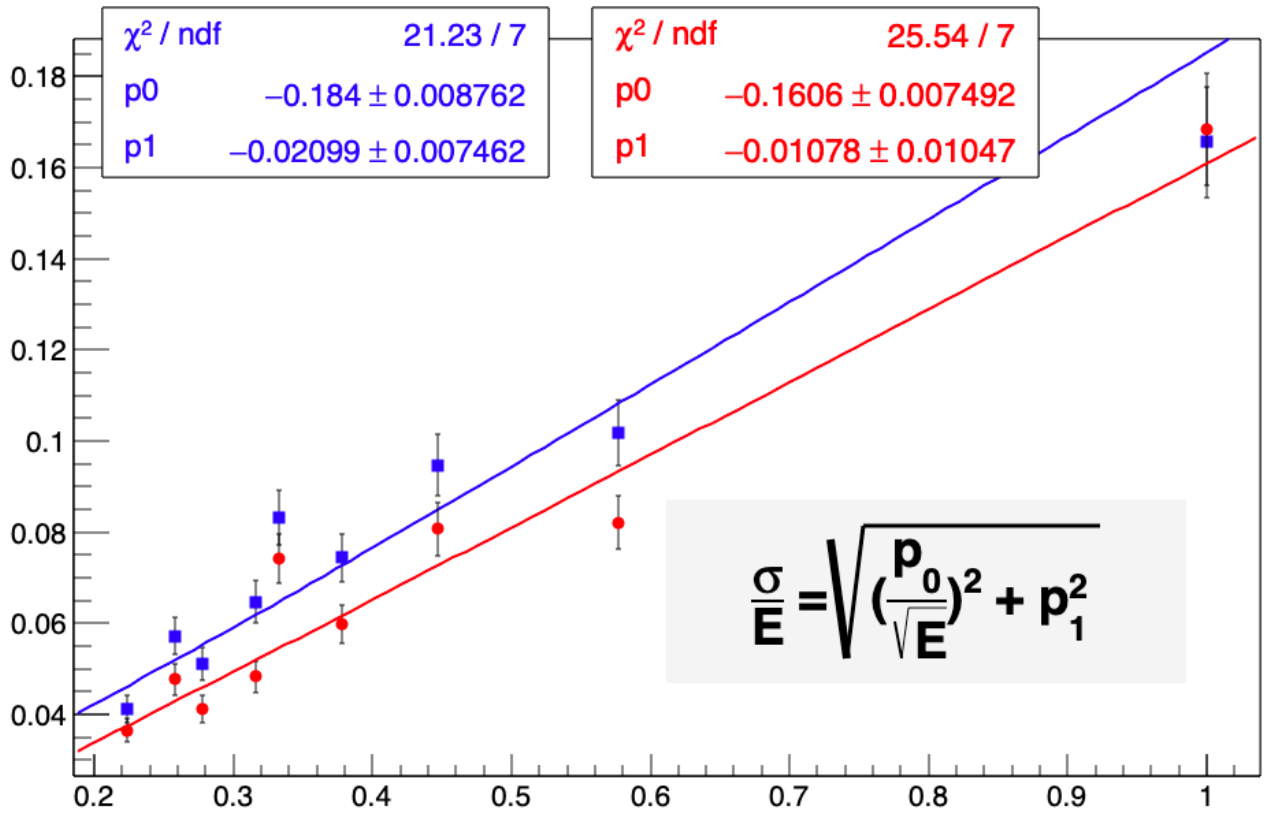








The energy resolution



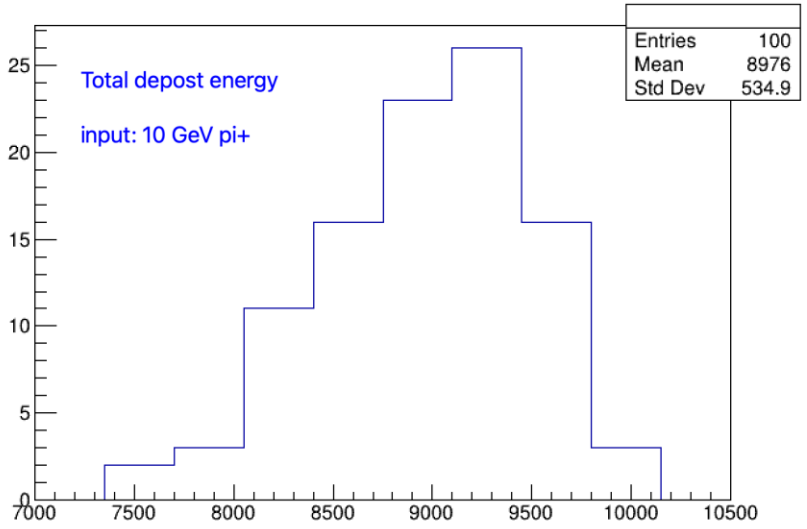
$$\frac{\sigma}{E} = \sqrt{\left(\frac{p_0}{\sqrt{E}}\right)^2 + p_1^2}$$

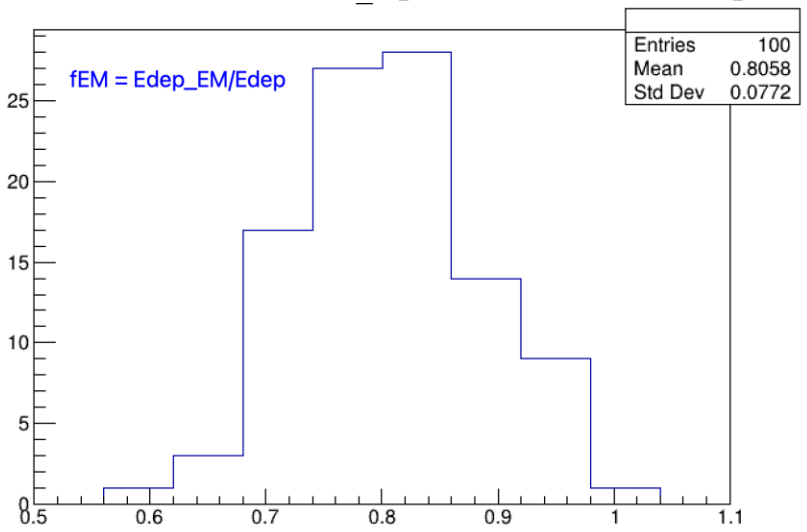
- Deposited energy after vs before corrected by the correlation.

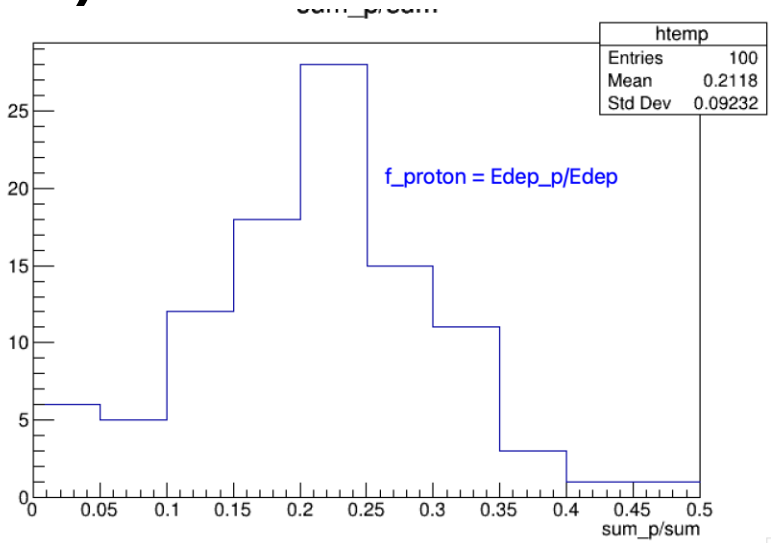
What is ongoing

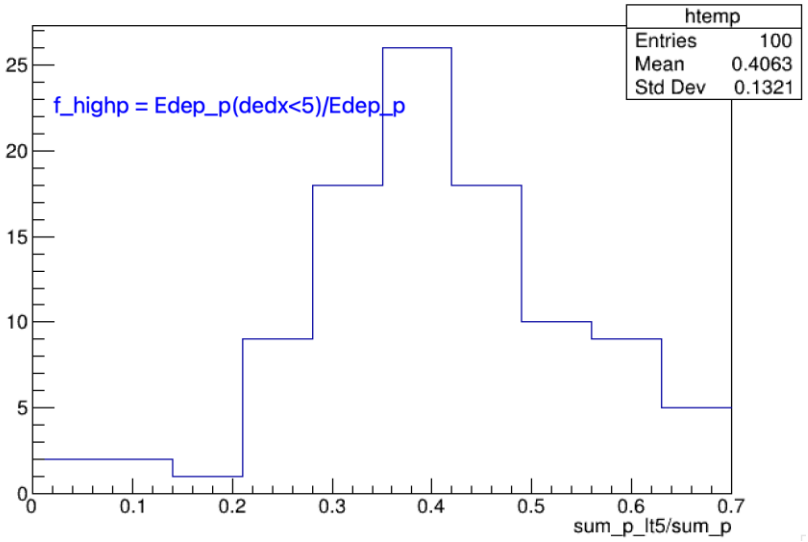
- Considering the time constants from proton measurements
  - Assume the fraction of fast component will be smaller than α and larger than γ ⇒ samples are ready (1-20 GeV, BGO)
  - Test beam with ZnWO4 and BGO crystal
    - BGO: Turns out the constants and fraction (proton) is consistent with those from electrons ⇒ not possible to extract accurate fEM from fit to time.
    - ZnWO4: time shape from high/low energy proton can be distinguished from electrons and α (pars from paper) ⇒ generating the samples with ZnWO4

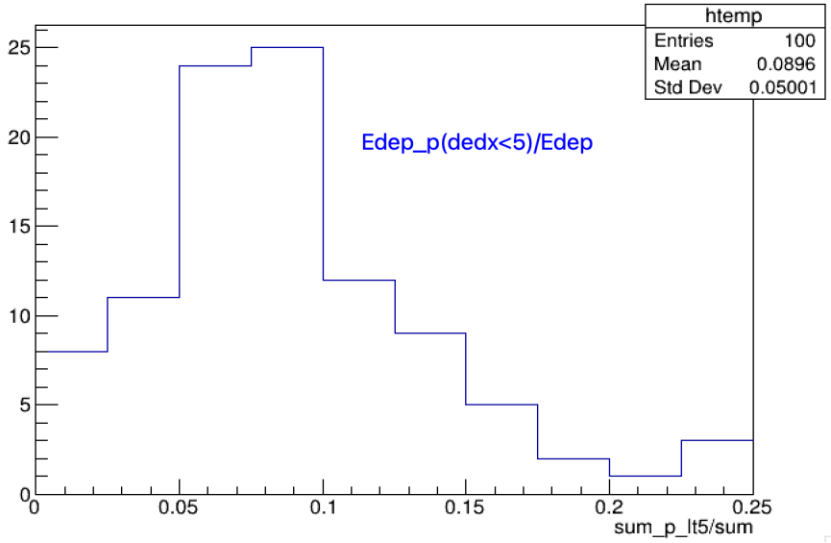
Sample checks (BGO + proton pars)

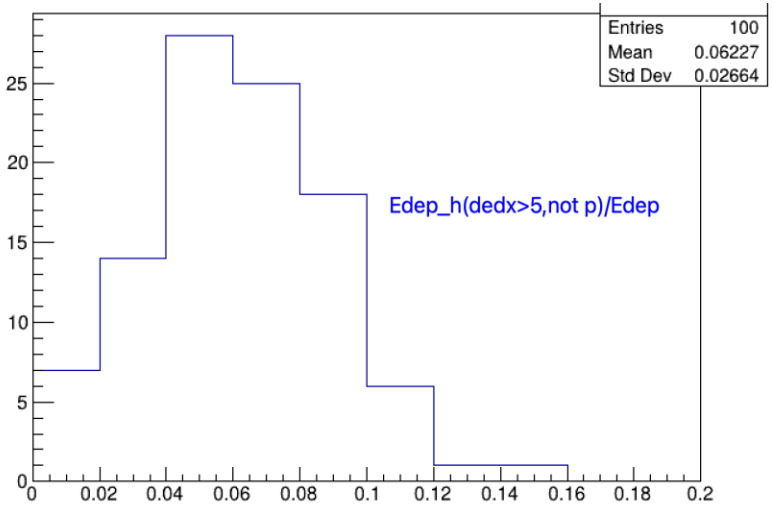




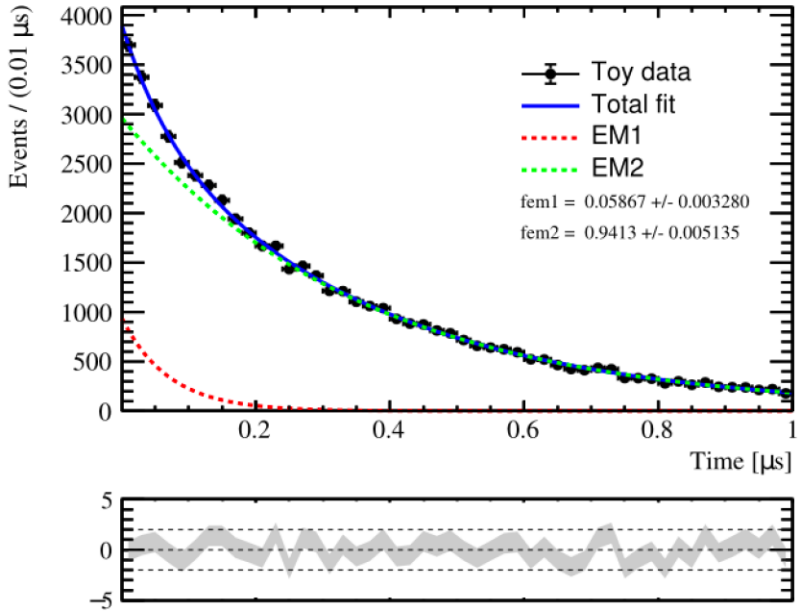


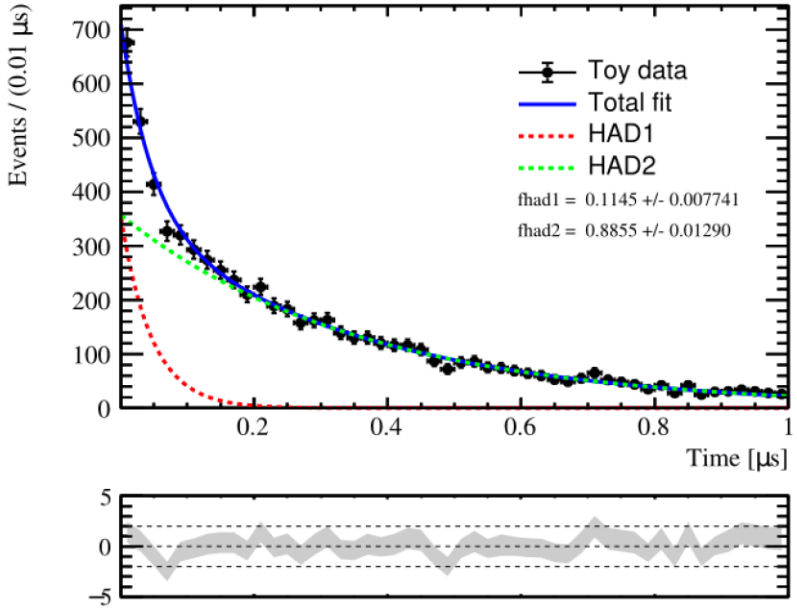


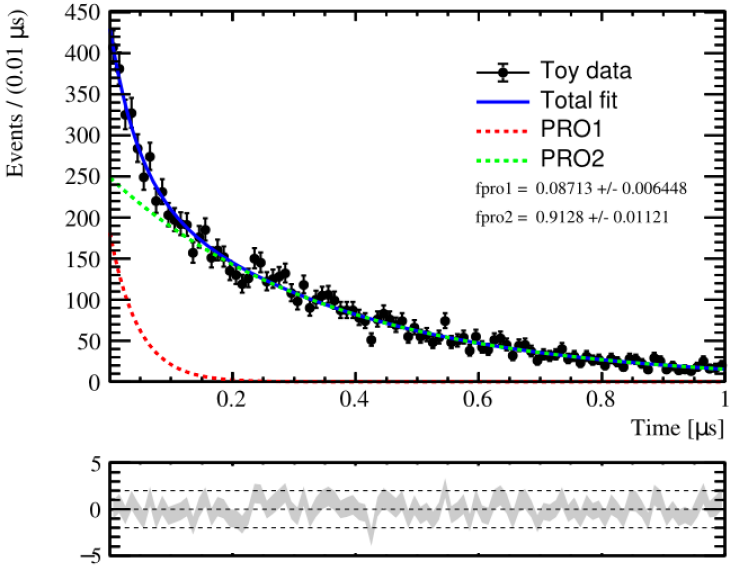




Fit to different components



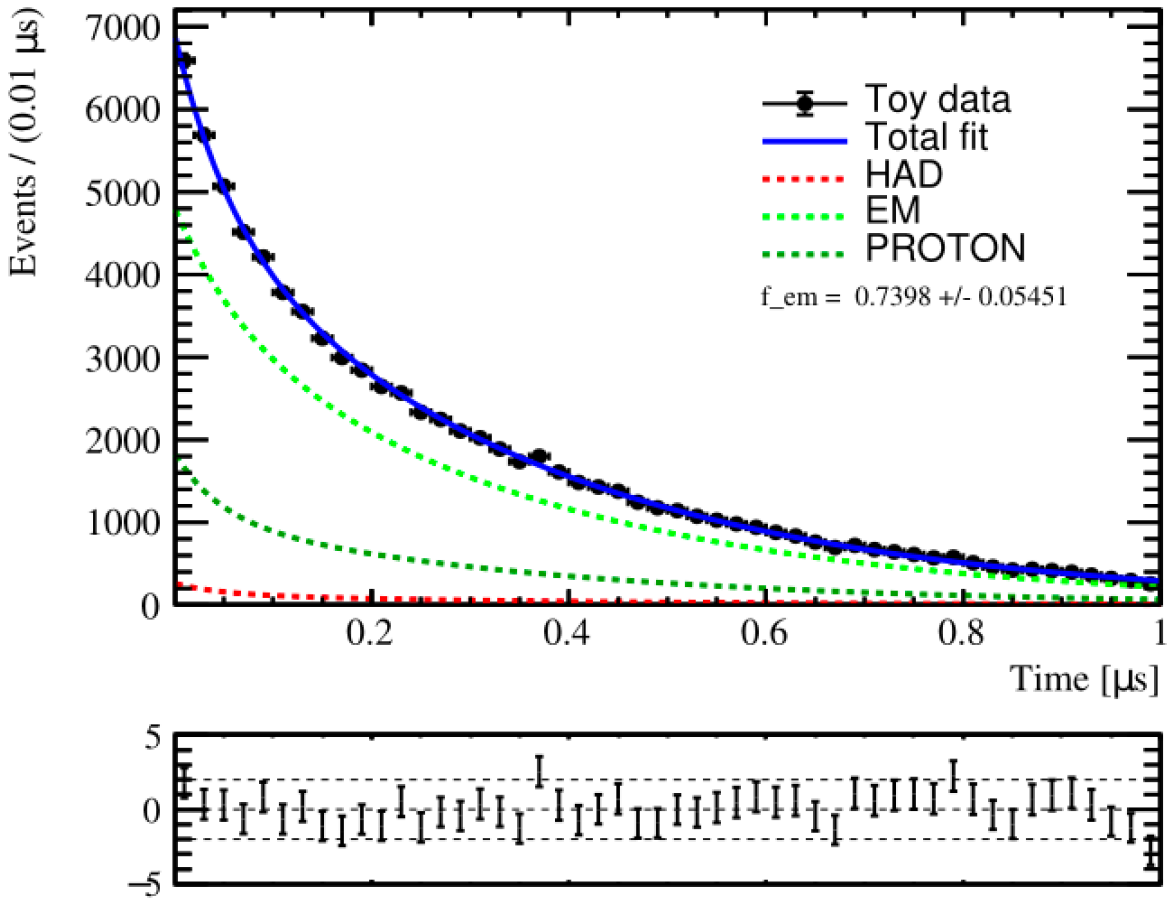




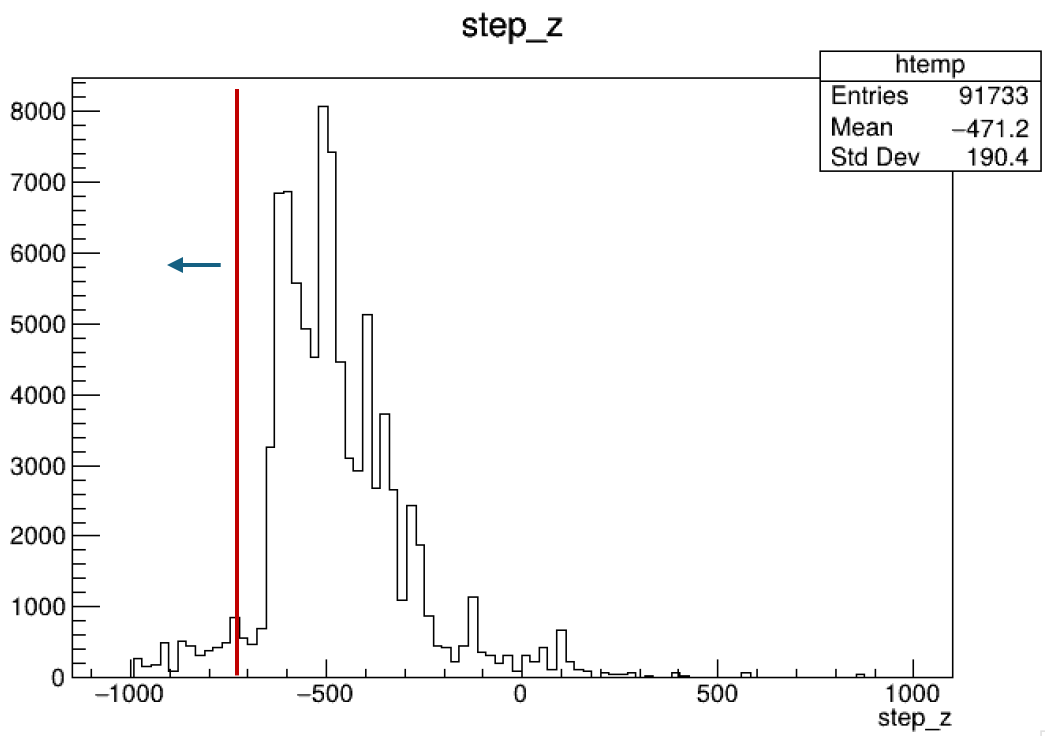
- Fraction_EM: 0.06 + 0.94
- Fraction_HAD: 0.12 + 0.88
- Fraction_PRO: 0.09 + 0.91

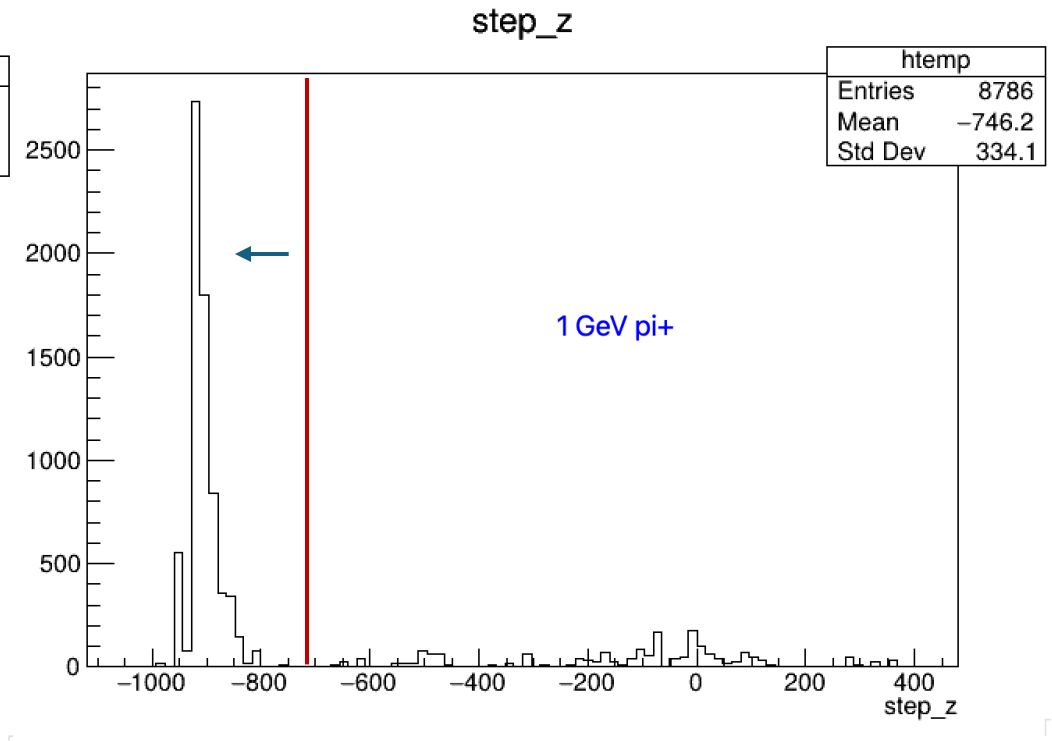
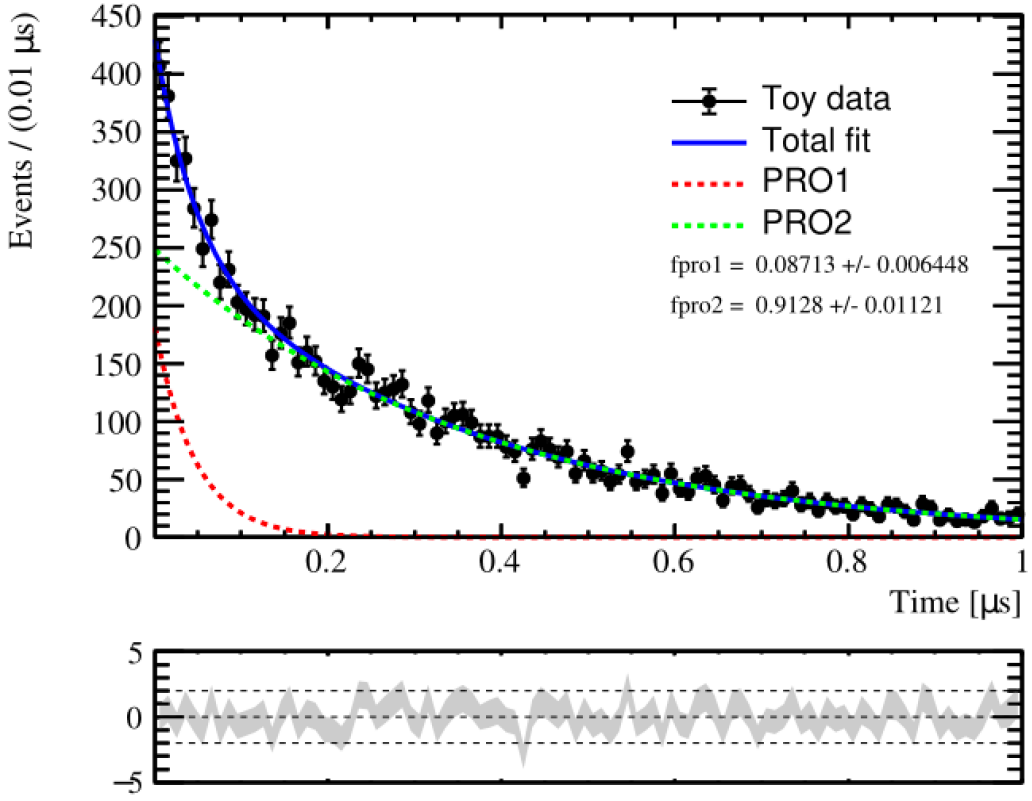
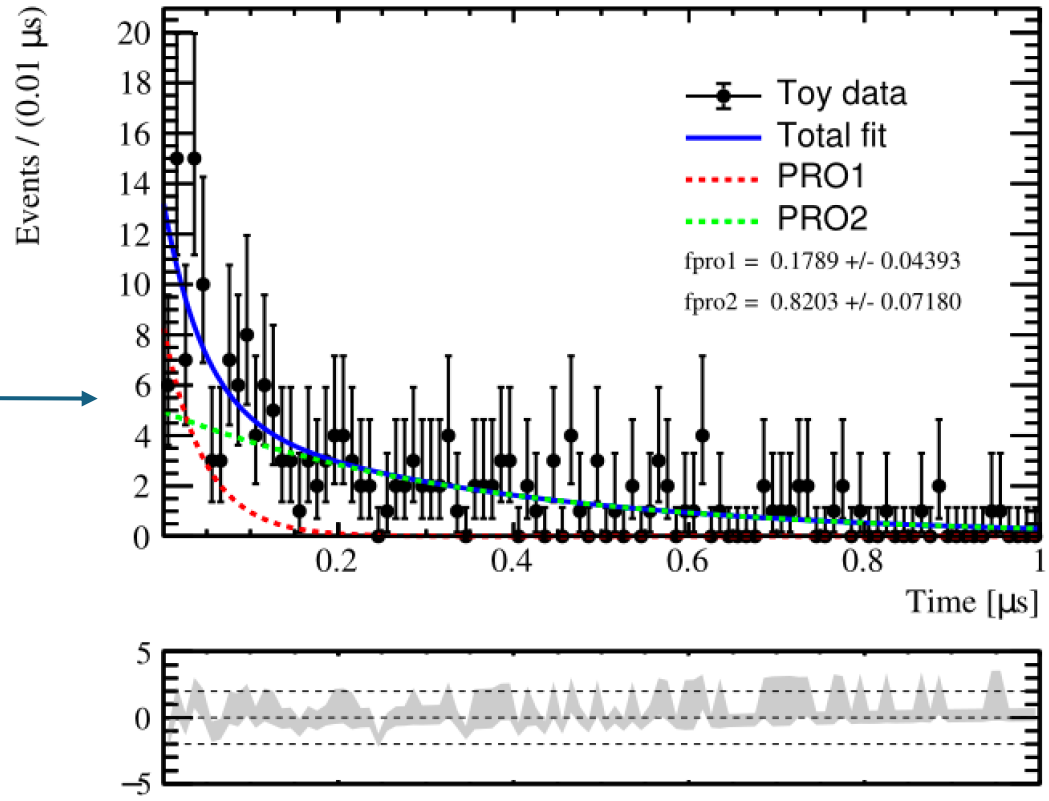
Fit to all scintillating photons

- F_em = 0.740+/-0.055 (0.738)
- F_had = 0.029+/-0.077(0.063)
- F_pro = 0.228+/-0.117(0.199)



With $24X_0$(26.9 cm for BGO) in z direction





Compare to full geometry